

OFFICE NETWORK BUILD

Static Routing

Backstory:

As the company continued to grow, it acquired an additional space within the same building that could be converted into a dedicated server room and IT infrastructure area. This provided an opportunity to move several important internal resources away from individual workstations and into a centralized environment.

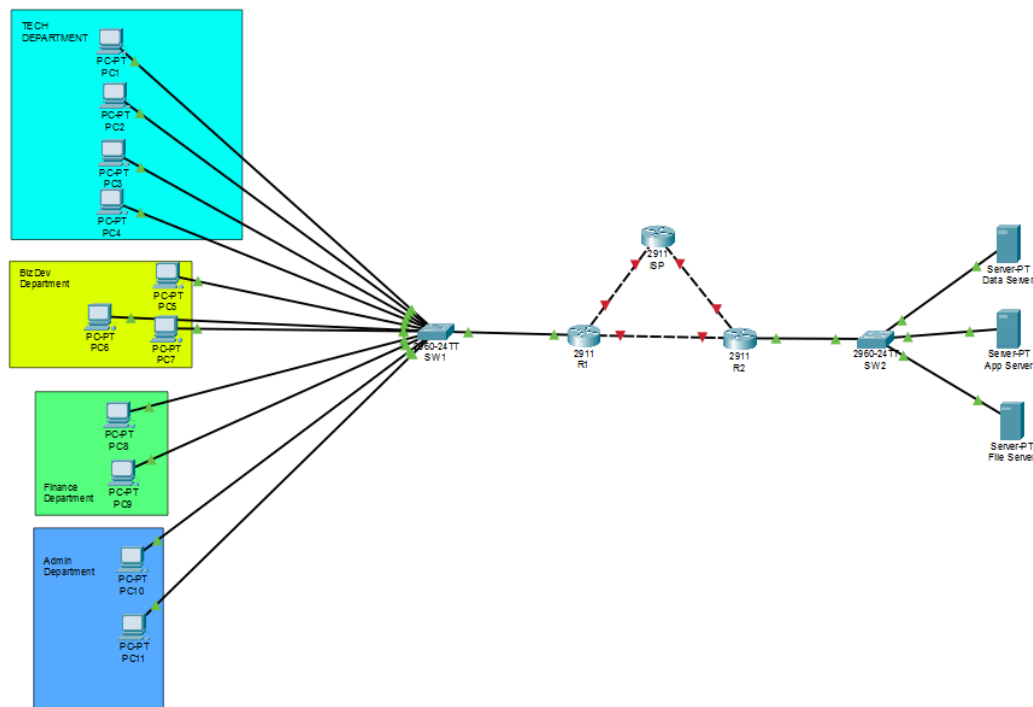
At the same time, the developers from the Tech team had created an internal inventory management system to help the company track its equipment and assets. The application needed a dedicated environment where it could be hosted and accessed by employees across different departments. The new server room therefore became the designated location for the inventory system and its supporting infrastructure.

The company's Disaster Risk Reduction monitoring portals also continued to operate through cloud infrastructure. However, to provide an additional copy of important monitoring data, the company decided to store selected portal data locally in addition to the existing cloud environment. This created a local data repository that could support backup, internal access, and future data processing requirements.

Finally, departments began centralizing their commonly used documents and company files into a local file server, reducing reliance on individual computers for shared files and providing a more organized way to manage departmental data.

With these changes, the network now needed to support a new Server Network separate from the existing departmental networks. The main office users would need controlled access to centralized resources such as the inventory system, locally stored monitoring data, and shared departmental files. This introduced the need for routing between the existing office network and the newly established server network.

Current Topology:



In the previous section, trunking and inter-VLAN routing were implemented to allow communication between the different departmental VLANs within the main office network. This provided the connectivity needed for employees to access resources beyond their own department while maintaining the network's logical segmentation.

As the company continued to develop its infrastructure, a dedicated server room was established within the same building to centralize important resources. The new server network now hosts the company's inventory system developed by the Tech team, locally stored data from the Disaster Risk Reduction monitoring portals, and shared files for the different departments.

Because the server network is a separate IP network from the main office network, the routers must be configured to know how to reach it. This section introduces static routing, where specific routes are manually configured to allow traffic between the main office network and the newly established server network.

IP Address Table:

Office Network Build					
Device	IP	Subnet	Default Gateway	VLAN	Interface
R1	192.168.100.1	255.255.255.248 - /29	N/A	VLAN 10	G0/0.10
R1	192.168.100.9	255.255.255.248 - /29	N/A	VLAN 20	G0/0.20
R1	192.168.100.17	255.255.255.248 - /29	N/A	VLAN 30	G0/0.30
R1	192.168.100.25	255.255.255.248 - /29	N/A	VLAN 40	G0/0.40
R1 (Transit Link)	192.168.102.1	255.255.255.252 - /30	N/A	N/A	G0/1
R2	192.168.101.1	255.255.255.0 - /24	N/A	N/A	G0/0
R2 (Transit Link)	192.168.102.2	255.255.255.252 - /30	N/A	N/A	G0/1
PC1	192.168.100.2	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC2	192.168.100.3	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC3	192.168.100.4	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC4	192.168.100.5	255.255.255.248 - /29	192.168.100.1	VLAN 10	Fa0
PC5	192.168.100.10	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC6	192.168.100.11	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC7	192.168.100.12	255.255.255.248 - /29	192.168.100.9	VLAN 20	Fa0
PC8	192.168.100.18	255.255.255.248 - /29	192.168.100.17	VLAN 30	Fa0
PC9	192.168.100.19	255.255.255.248 - /29	192.168.100.17	VLAN 30	Fa0
PC10	192.168.100.26	255.255.255.248 - /29	192.168.100.25	VLAN 40	Fa0
PC11	192.168.100.27	255.255.255.248 - /29	192.168.100.25	VLAN 40	Fa0
Data Server	192.168.101.10	255.255.255.0 - /24	192.168.101.1	N/A	Fa0
Application Server	192.168.101.20	255.255.255.0 - /24	192.168.101.1	N/A	Fa0
File Server	192.168.101.30	255.255.255.0 - /24	192.168.101.1	N/A	Fa0

The image above shows the IP Addressing and VLAN Assignment Table for the Office and Server Networks. This table summarizes the IP addresses, subnet masks, default gateways, VLAN assignments, and interfaces used across the main office network and the newly established server network, including the R1–R2 transit link.

```

R1
R1#
R1#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset    up          up
GigabitEthernet0/0.10 192.168.100.1   YES manual    up          up
GigabitEthernet0/0.20 192.168.100.9   YES manual    up          up
GigabitEthernet0/0.30 192.168.100.17  YES manual    up          up
GigabitEthernet0/0.40 192.168.100.25  YES manual    up          up
GigabitEthernet0/1     192.168.102.1   YES manual    up          up
GigabitEthernet0/2     unassigned      YES unset    up          down
Vlan1               unassigned      YES unset    administratively down down
R1#
  
```

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R2

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

R2>
R2>
R2>en
R2#show ip int brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.101.1	YES	manual	up	up
GigabitEthernet0/1	192.168.102.2	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	down
Vlan1	unassigned	YES	unset	administratively down	down

R2#

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Ping Test of Transit Links:

R1

```
R1>  
R1>ping ?  
WORD Ping destination address or hostname  
ip IP echo  
ipv6 IPv6 echo  
R1>ping 192.168.102.2
```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.102.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R1>

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R2

```
R2>  
R2>ping 192.168.102.1
```

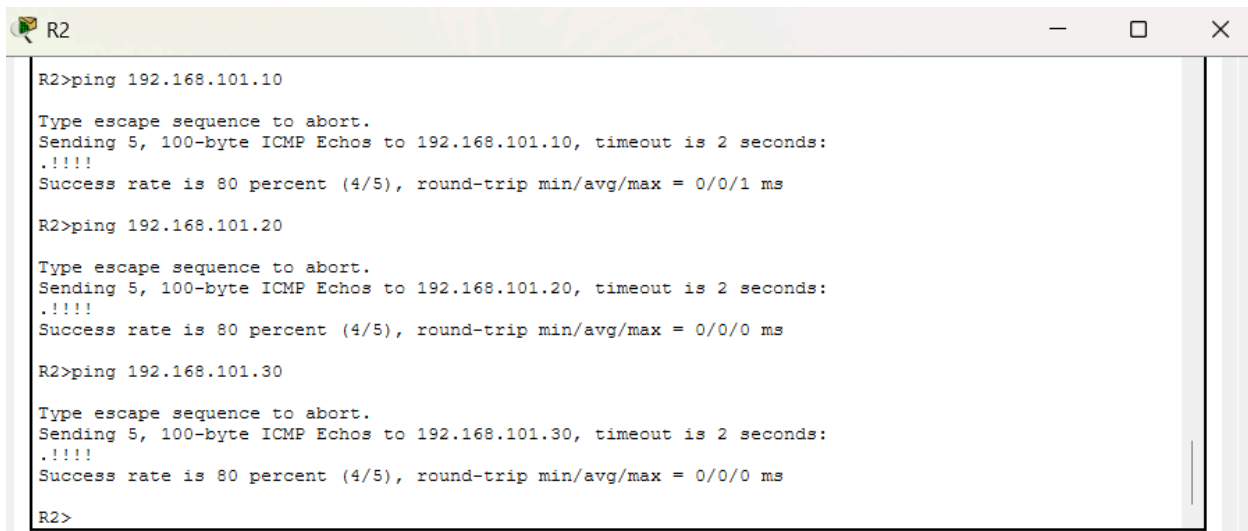
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.102.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R2>

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Verifying Servers can be reached from R2:



A screenshot of a Cisco Packet Tracer window titled 'R2'. The window contains a command prompt interface with the following text:

```
R2>ping 192.168.101.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.101.10, timeout is 2 seconds:
.....
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

R2>ping 192.168.101.20

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.101.20, timeout is 2 seconds:
.....
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

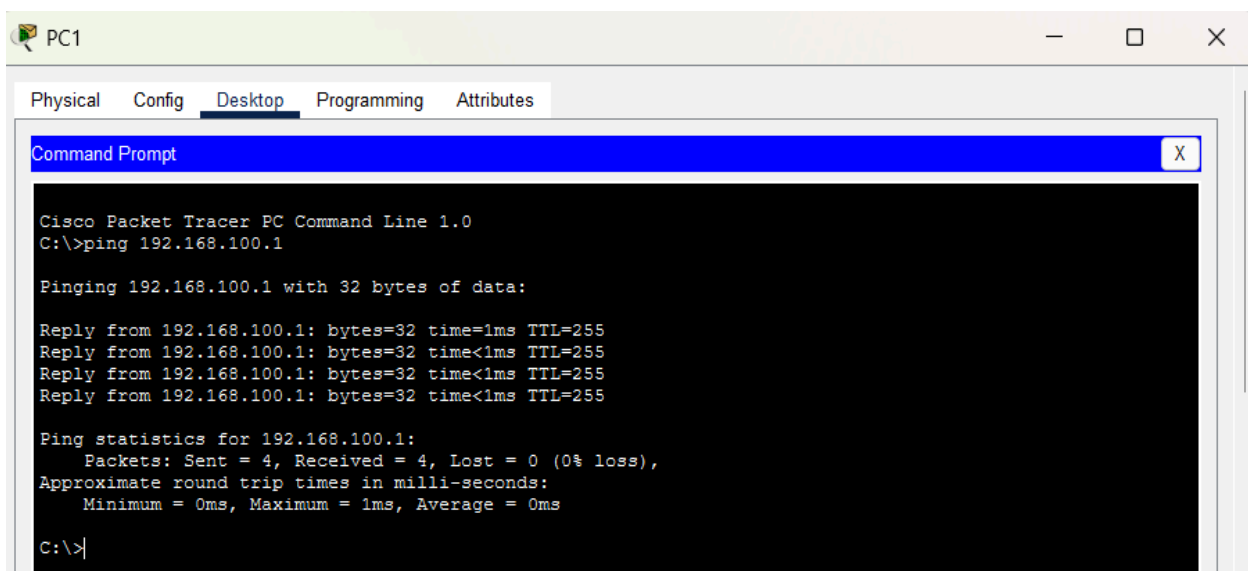
R2>ping 192.168.101.30

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.101.30, timeout is 2 seconds:
.....
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R2>
```

PCs try to reach their gateway:

Tech PC:



A screenshot of a Cisco Packet Tracer window titled 'PC1'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, showing a 'Command Prompt' window. The Command Prompt contains the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.100.1

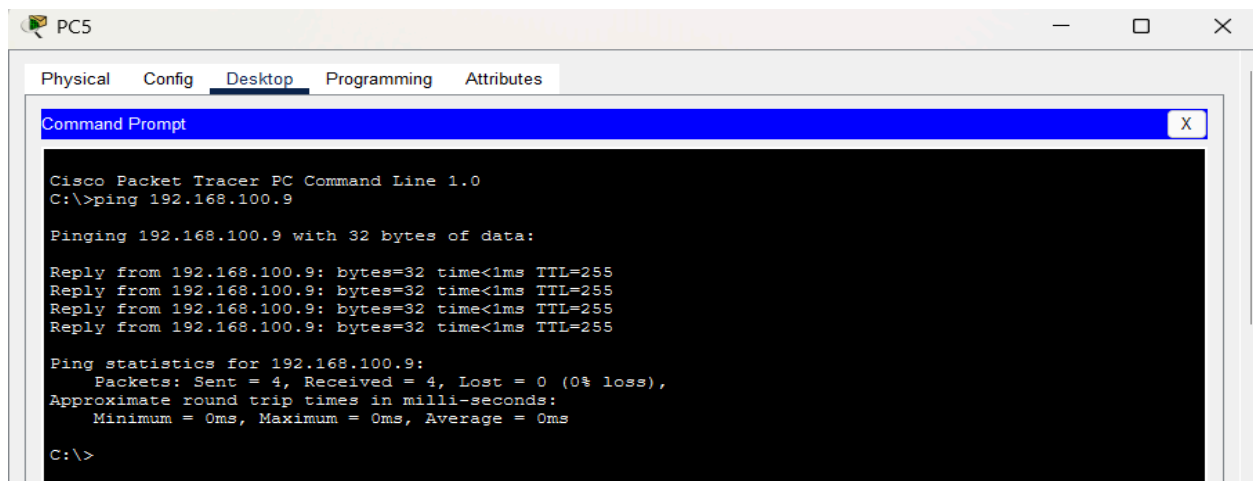
Pinging 192.168.100.1 with 32 bytes of data:

Reply from 192.168.100.1: bytes=32 time=1ms TTL=255
Reply from 192.168.100.1: bytes=32 time<1ms TTL=255
Reply from 192.168.100.1: bytes=32 time<1ms TTL=255
Reply from 192.168.100.1: bytes=32 time<1ms TTL=255

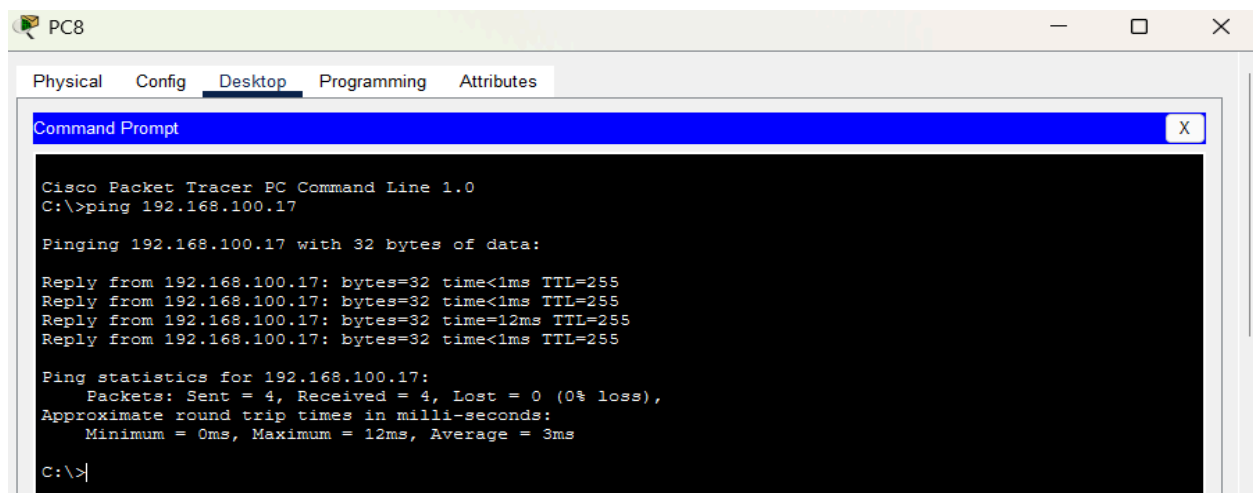
Ping statistics for 192.168.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

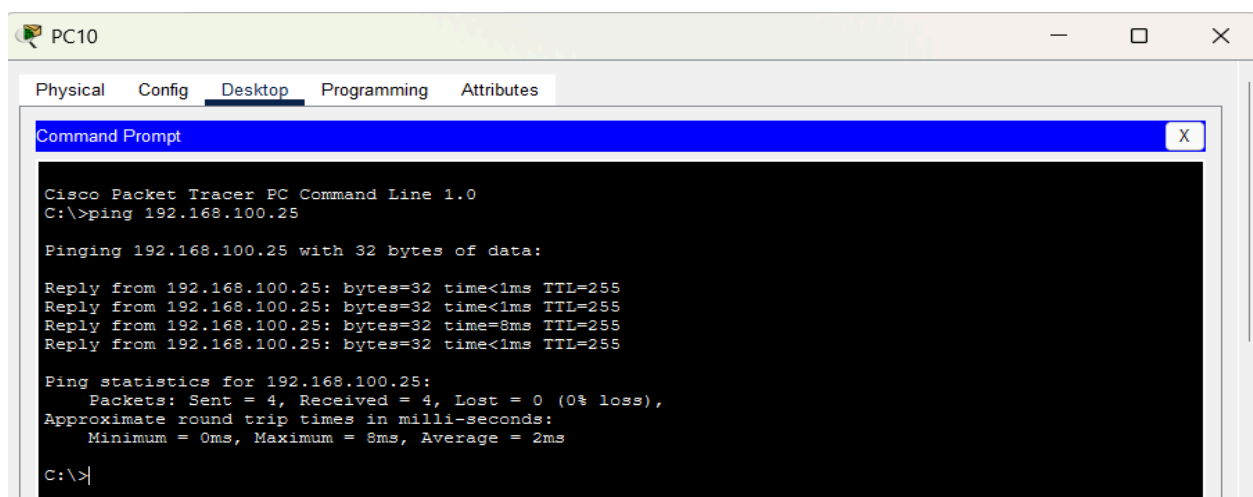
BizDev PC:



Finance PC:

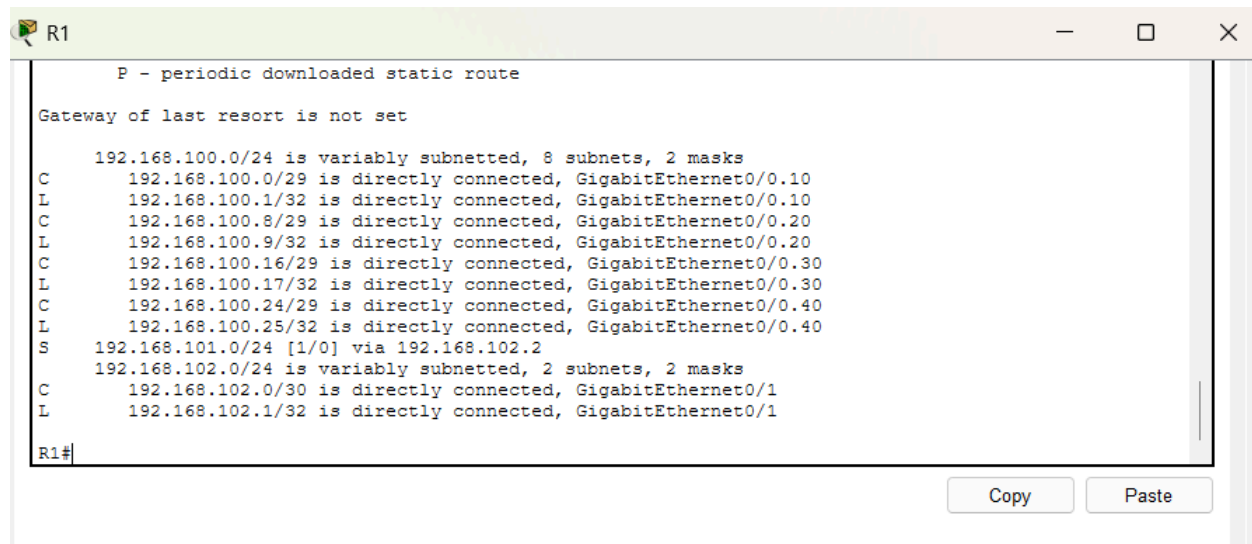


Admin PC:



Now that the transit links are working, we can proceed with Static Routing.

So we configured a static route in R1 so that it can reach the network 192.168.101.0/24 via the Transit Link Address on R2, 192.168.102.2.



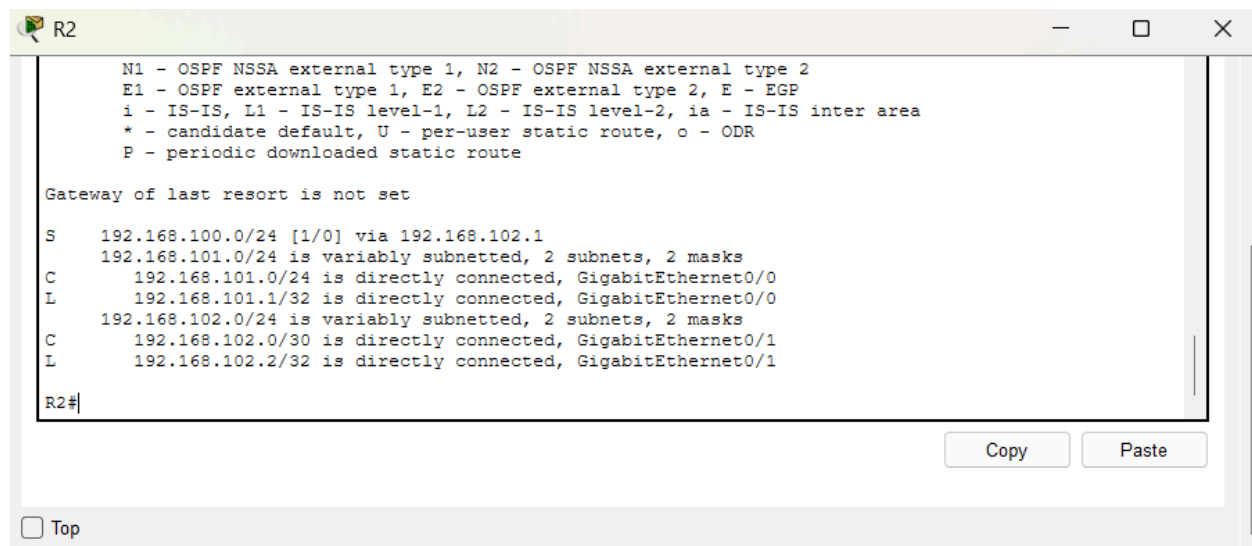
```
R1
P - periodic downloaded static route

Gateway of last resort is not set

  192.168.100.0/24 is variably subnetted, 8 subnets, 2 masks
C    192.168.100.0/29 is directly connected, GigabitEthernet0/0.10
L    192.168.100.1/32 is directly connected, GigabitEthernet0/0.10
C    192.168.100.8/29 is directly connected, GigabitEthernet0/0.20
L    192.168.100.9/32 is directly connected, GigabitEthernet0/0.20
C    192.168.100.16/29 is directly connected, GigabitEthernet0/0.30
L    192.168.100.17/32 is directly connected, GigabitEthernet0/0.30
C    192.168.100.24/29 is directly connected, GigabitEthernet0/0.40
L    192.168.100.25/32 is directly connected, GigabitEthernet0/0.40
S    192.168.101.0/24 [1/0] via 192.168.102.2
  192.168.102.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.102.0/30 is directly connected, GigabitEthernet0/1
L    192.168.102.1/32 is directly connected, GigabitEthernet0/1

R1#
```

And we configured a static route in R2 as well so that it can reach the network 192.168.100.0/24 via the Transit Link Address on R1, 192.168.102.1.



```
R2
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

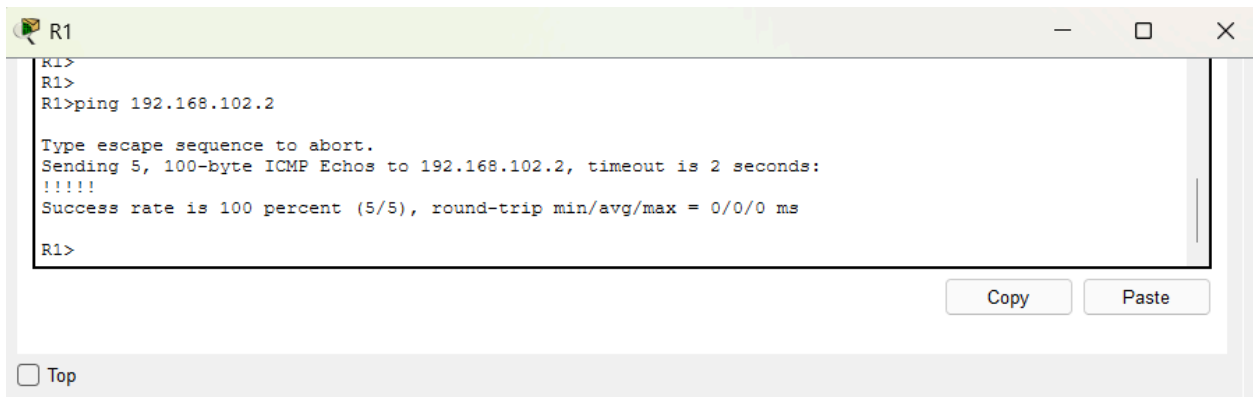
S    192.168.100.0/24 [1/0] via 192.168.102.1
  192.168.101.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.101.0/24 is directly connected, GigabitEthernet0/0
L    192.168.101.1/32 is directly connected, GigabitEthernet0/0
  192.168.102.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.102.0/30 is directly connected, GigabitEthernet0/1
L    192.168.102.2/32 is directly connected, GigabitEthernet0/1

R2#
```

***Note:** We configured the static route to 192.168.100.0/24 instead of /29 so that it would include all the other subnets in the static route.

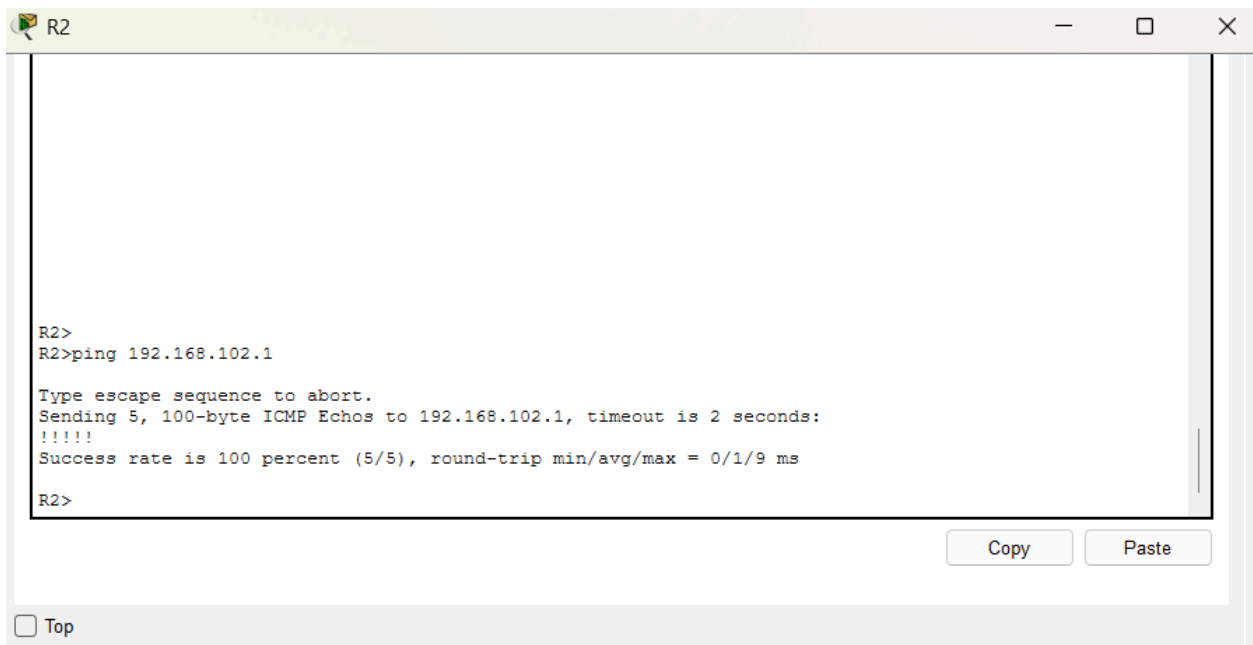
Testing of Static Routes

Testing Transit Links:



```
R1>  
R1>  
R1>ping 192.168.102.2  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.102.2, timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms  
  
R1>
```

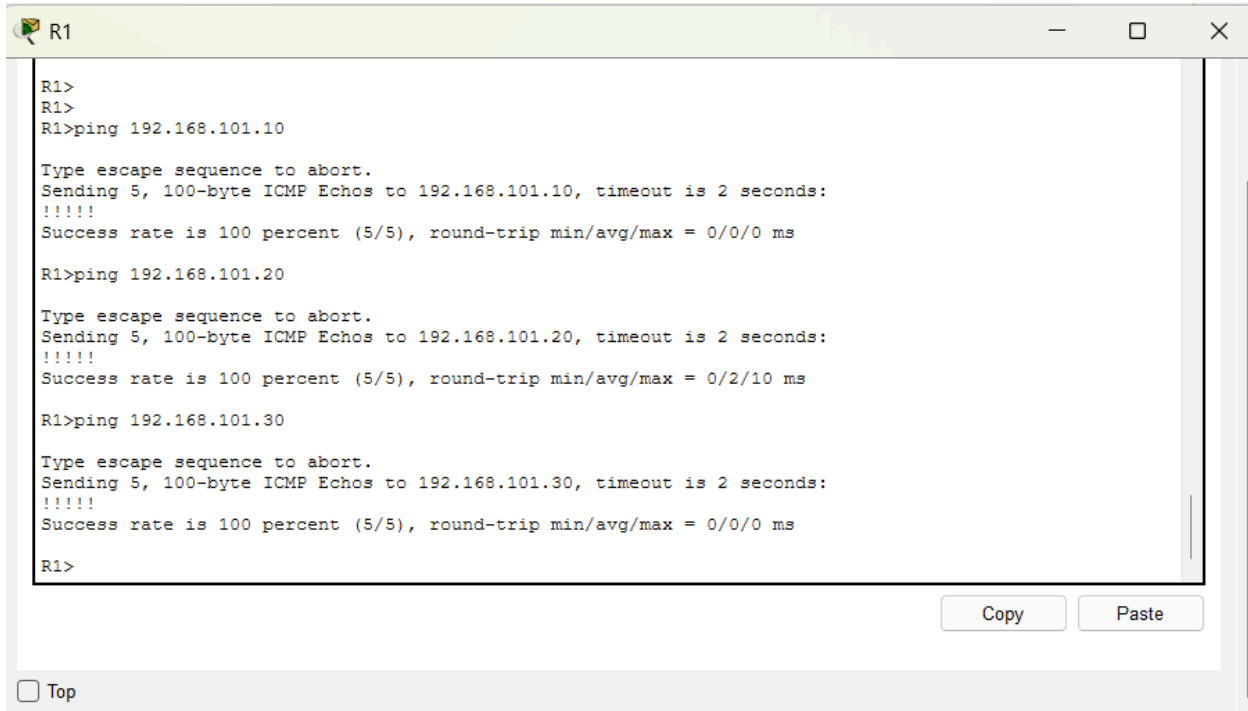
☐ Top



```
R2>  
R2>  
R2>ping 192.168.102.1  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.102.1, timeout is 2 seconds:  
!!!!!  
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/9 ms  
  
R2>
```

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Test R1 reaching servers in 192.168.101.0/24 Network (R1 to Servers):



R1

```
R1>
R1>
R1>ping 192.168.101.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.101.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1>ping 192.168.101.20

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.101.20, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/10 ms

R1>ping 192.168.101.30

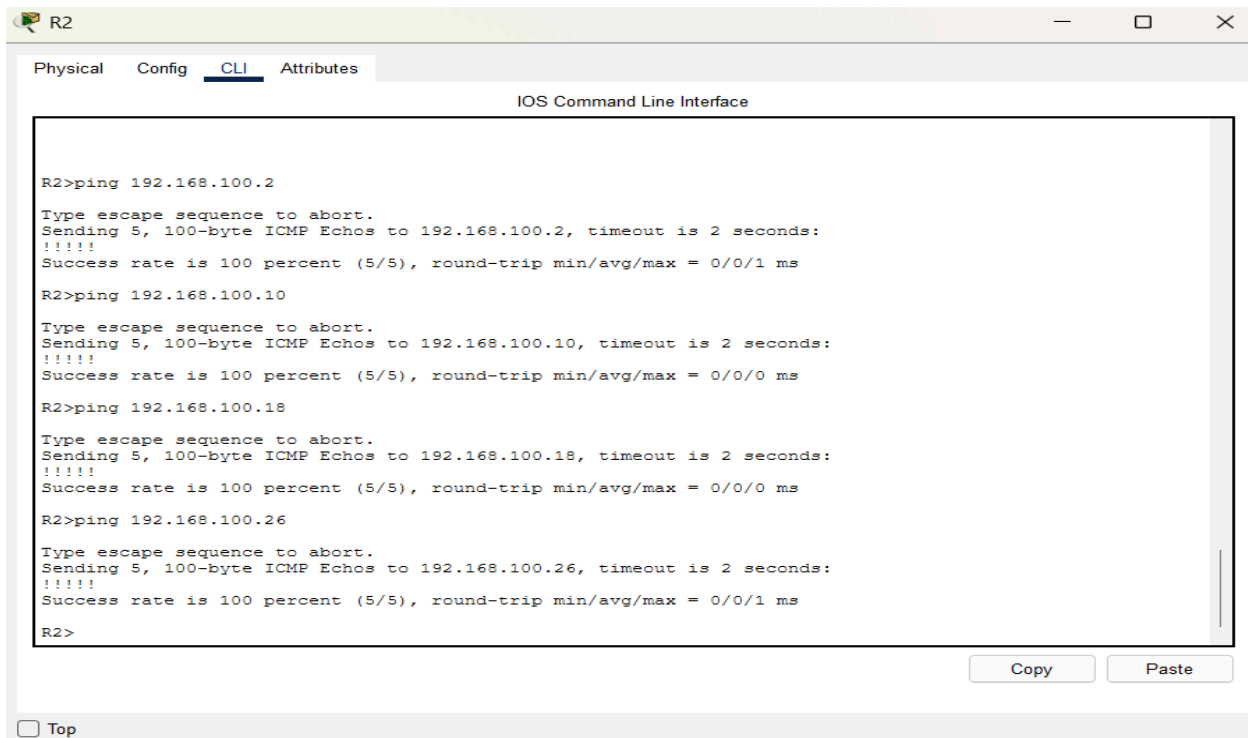
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.101.30, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1>
```

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Test R2 reaching PCs in 192.168.100.0/24 Network (R2 to Office Network):



R2

Physical Config CLI Attributes

IOS Command Line Interface

```
R2>ping 192.168.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R2>ping 192.168.100.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R2>ping 192.168.100.18

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.18, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R2>ping 192.168.100.26

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.26, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

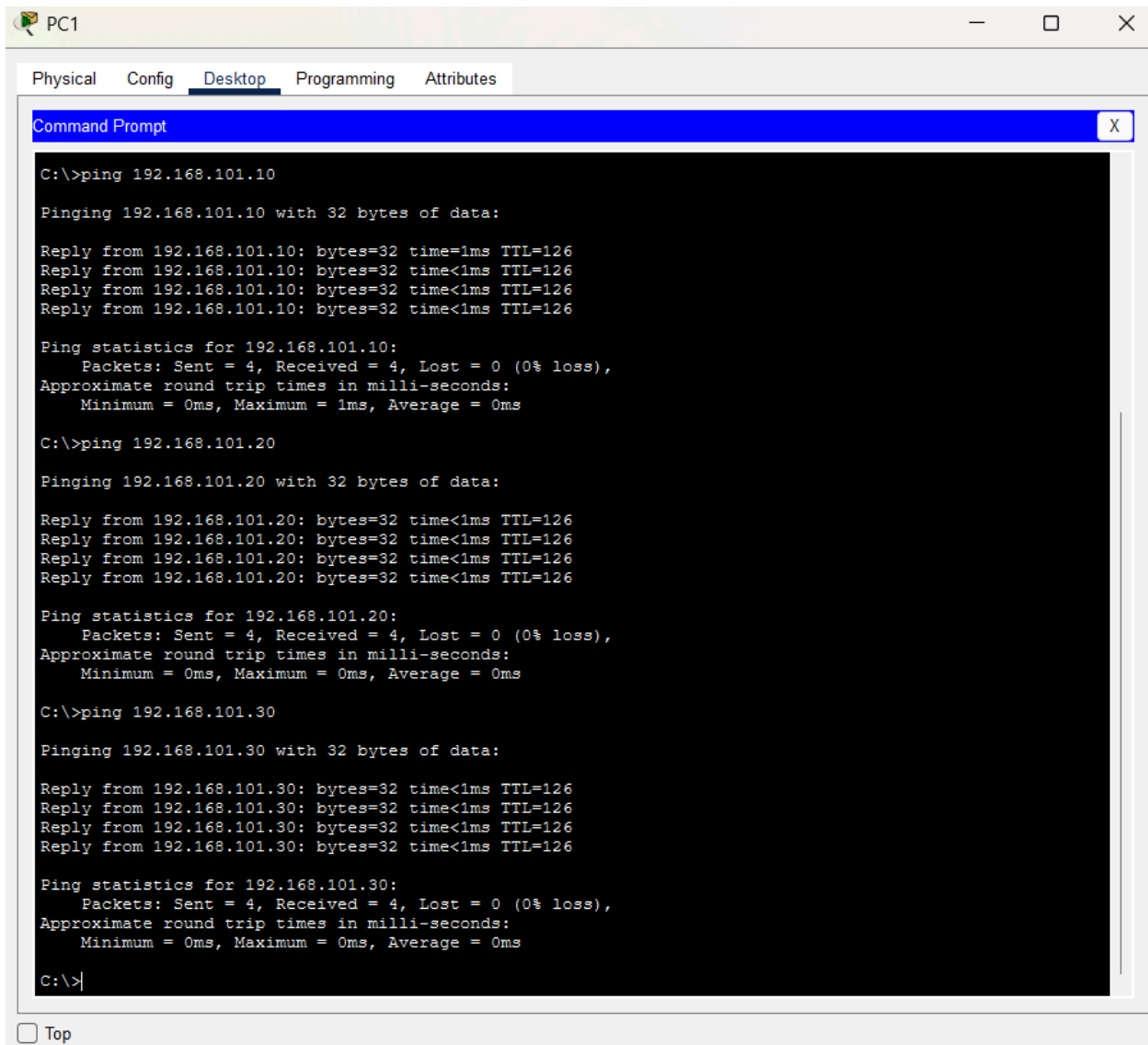
R2>
```

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Testing PC connectivity with Servers:

Tech PC:



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.101.10

Pinging 192.168.101.10 with 32 bytes of data:

Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.101.20

Pinging 192.168.101.20 with 32 bytes of data:

Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

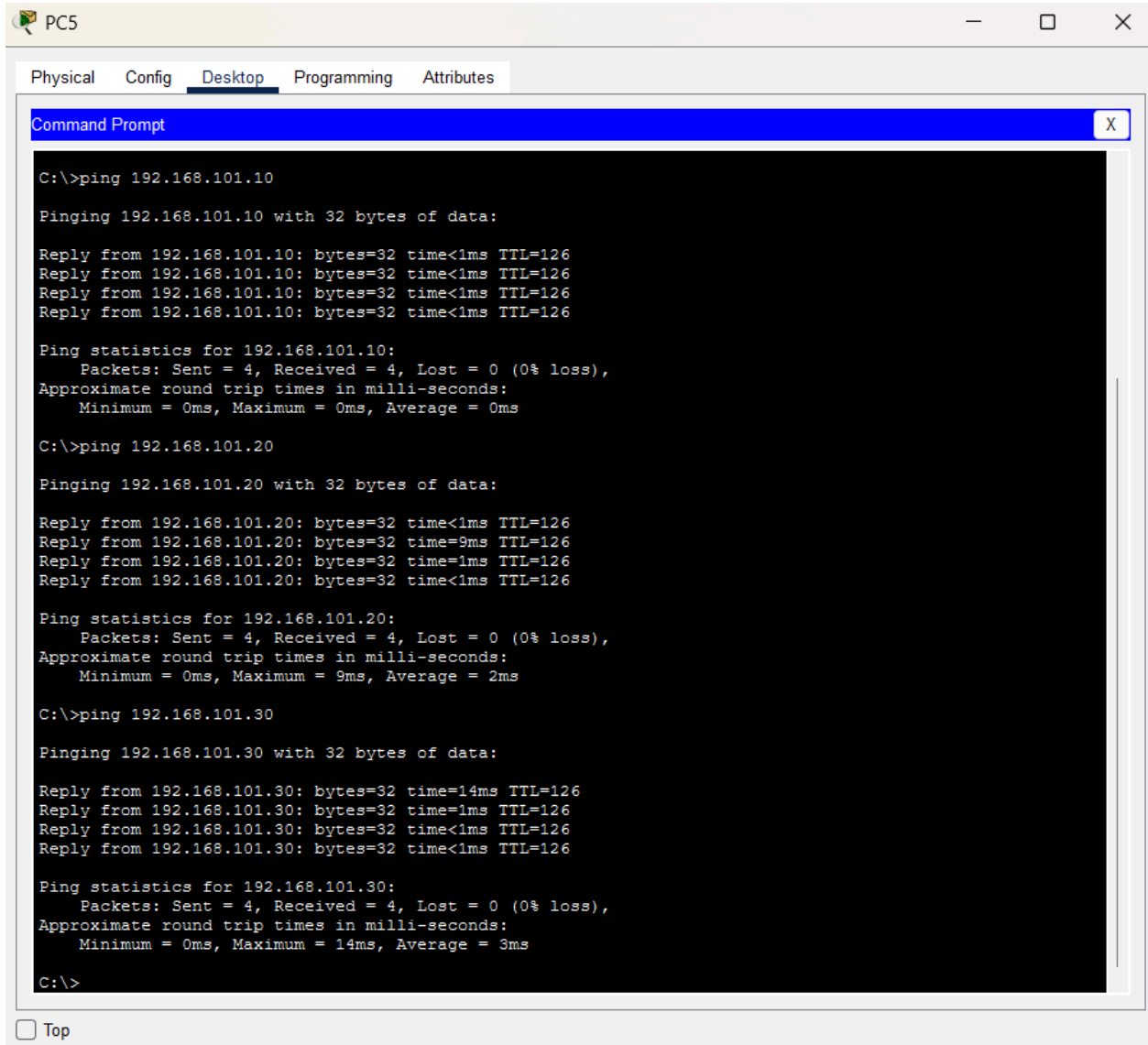
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

☐ Top

BizDev PC:



PC5

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.101.10

Pinging 192.168.101.10 with 32 bytes of data:

Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.20

Pinging 192.168.101.20 with 32 bytes of data:

Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time=9ms TTL=126
Reply from 192.168.101.20: bytes=32 time=1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

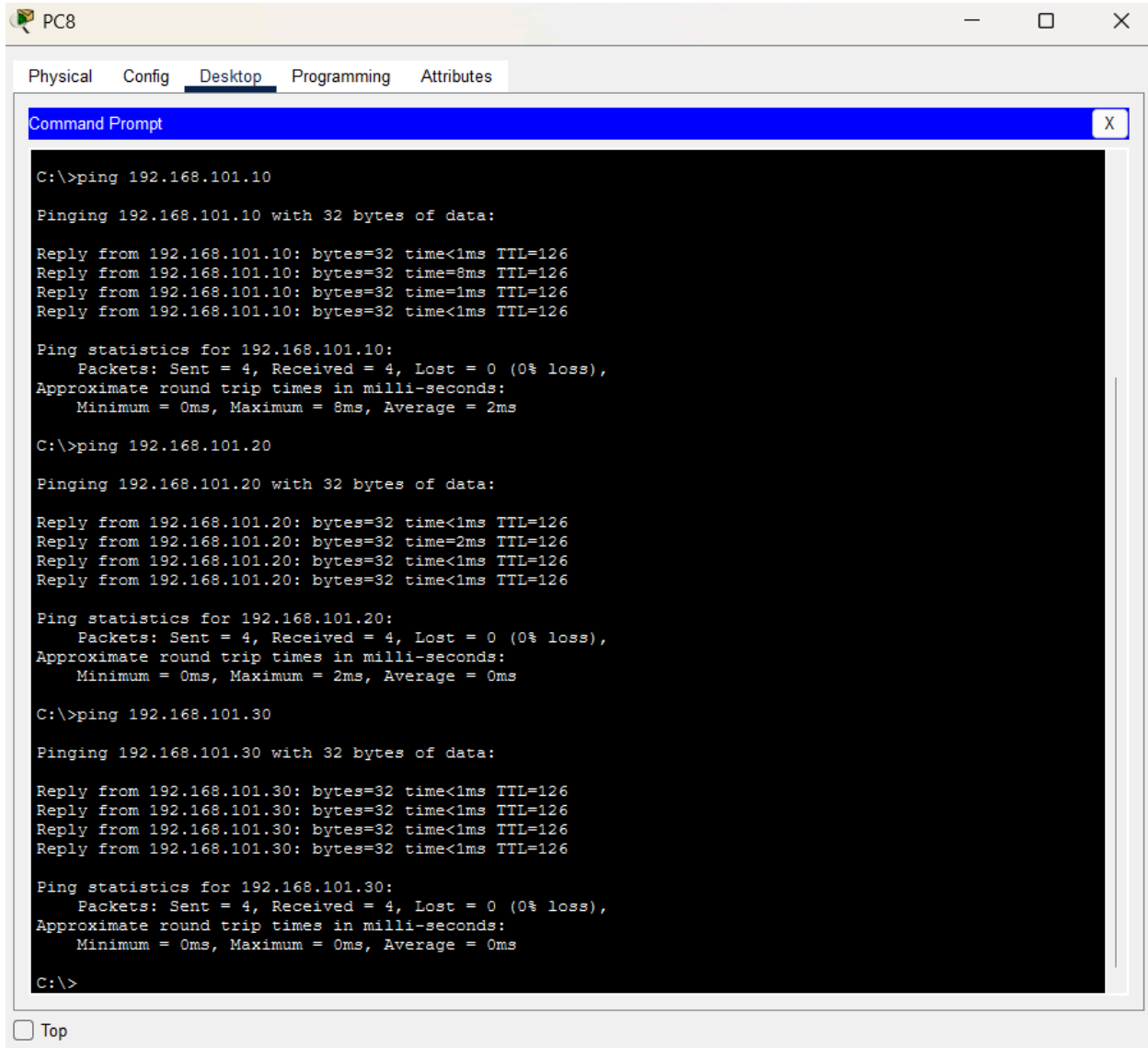
Reply from 192.168.101.30: bytes=32 time=14ms TTL=126
Reply from 192.168.101.30: bytes=32 time=1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>
```

☐ Top

Finance PC:



The screenshot shows a window titled "PC8" with a menu bar containing "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the execution of three ping commands to IP addresses 192.168.101.10, 192.168.101.20, and 192.168.101.30. Each command is followed by four replies and a summary of ping statistics. The statistics for each IP show 4 packets sent and received, 0% loss, and round trip times in milliseconds.

```
C:\>ping 192.168.101.10

Pinging 192.168.101.10 with 32 bytes of data:

Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time=8ms TTL=126
Reply from 192.168.101.10: bytes=32 time=1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping 192.168.101.20

Pinging 192.168.101.20 with 32 bytes of data:

Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time=2ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

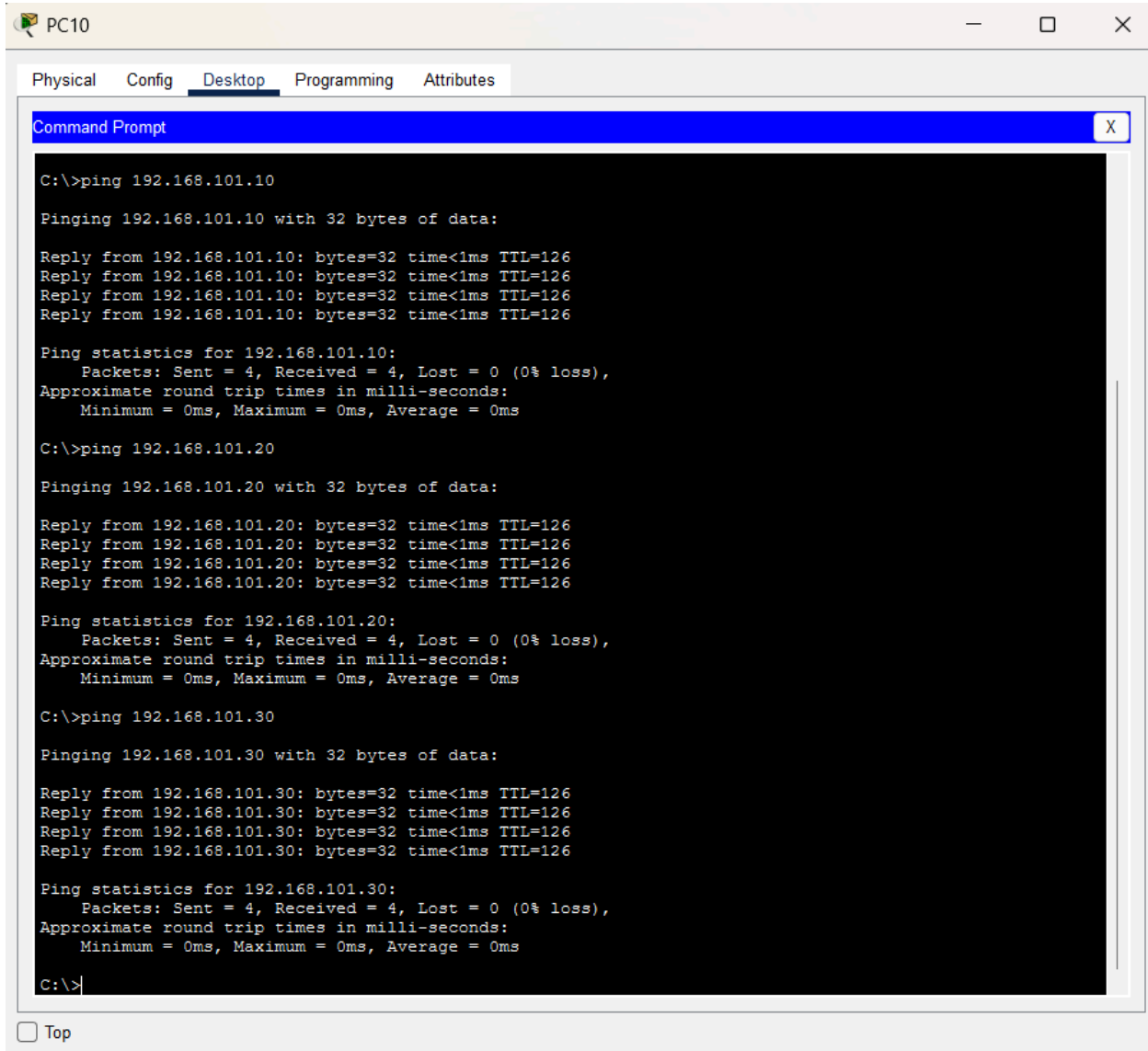
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

☐ Top

Admin PC:



PC10

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.101.10

Pinging 192.168.101.10 with 32 bytes of data:

Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126
Reply from 192.168.101.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.20

Pinging 192.168.101.20 with 32 bytes of data:

Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126
Reply from 192.168.101.20: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.101.30

Pinging 192.168.101.30 with 32 bytes of data:

Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126
Reply from 192.168.101.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.101.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

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Conclusion:

The implementation of static routing successfully extended the network beyond the main office and established connectivity with the newly created server network. The existing departmental VLANs and inter-VLAN routing provided communication within the main office, while the addition of a separate 192.168.101.0/24 server network introduced a new routing requirement. To connect these networks, a point-to-point transit network, 192.168.102.0/30, was configured between R1 and R2.

Static routes were then configured on both routers so that traffic could be forwarded between the main office and the server network. R1 was given a route to 192.168.101.0/24 through R2, while R2 was given a return route toward the main office network through R1. This demonstrated an important routing principle: routers must have a path to both the destination network and the return path for end-to-end communication to succeed.

The expanded topology also gave the network a practical purpose. The server network represents centralized company resources, including locally stored monitoring data, the internal inventory application, and shared departmental files. This allows users in the main office to access centralized resources without placing those services directly inside the individual departmental networks.

Through this implementation, the project demonstrated the progression from local network connectivity to routed network communication. VLANs provided segmentation, inter-VLAN routing enabled communication between departmental networks, and static routing extended that communication to a separate network containing centralized resources. This creates a more scalable foundation for the next stage of the project, where the network can be expanded further and dynamic routing can be introduced to reduce the need for manually maintained routes.